

Can Greenpoint get louder on paper?

notes for a possible model, using 2015 noise complaints

1 What I am trying to understand

I live in Greenpoint, so I wanted to use a place I know. I am asking whether the neighborhood could look louder in the city record even if the actual sound stayed the same. The thing I change is the chance that someone reports nighttime noise.

2 The dataset

NYC 311 Service Requests, collected by NYC311 and responding agencies. I filtered the archive to ZIP 11222, the year 2015, and complaint types beginning with Noise. The result is 2,425 geocoded rows. One row is one request, not one measured sound.

fields I use:

created date and hour - complaint type - descriptor - latitude - longitude

3 Research that changed the question

DOHMH shows that 311 complaint sources and survey-reported disturbance do not match. Kontokosta and Hong show that reporting is socially uneven. Gong et al. use Poisson models for NYC complaint counts. I simplify that idea into expected hourly counts.

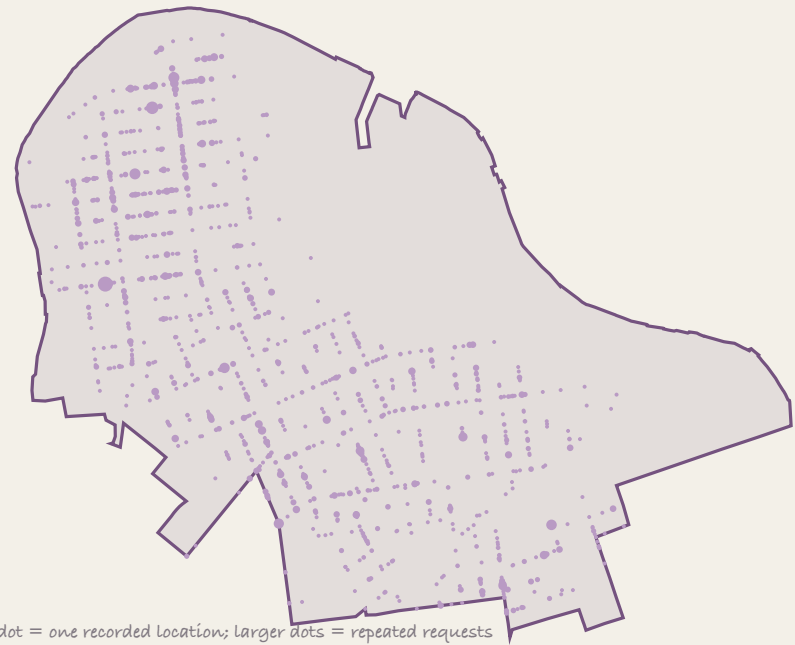
4 My starting record

actual 2015 row

ID 29607573 - 2015-01-01 00:28

Residential - loud music or party

the observed archive



each dot = one recorded location; larger dots = repeated requests

2,425 rows - 1,031 coordinate pairs - 8 complaint types

five largest complaint categories

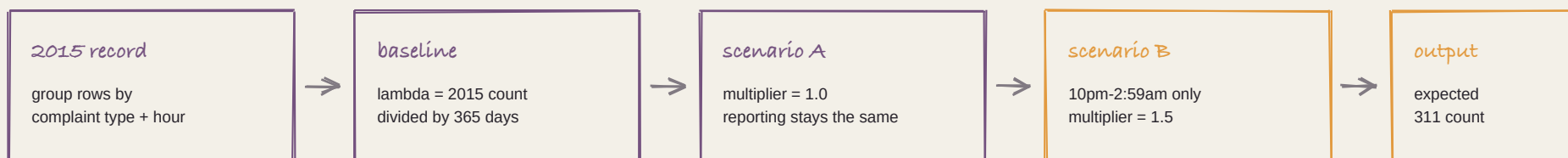
Residential	874
Unspecified	679
Commercial	388
Street - sidewalk	240
Vehicle	201

Important: this is a map of reports, not a map of decibels.

My rule: change reporting, not sound

a simple sensitivity test, not a claim about the real future

5 The rule



1.5 is my assumption and the number I am least certain about.

6 The run

I run the rule for one future week: June 3-9, 2030. That is 168 hourly steps. This first sketch is deterministic, so nothing is left to chance. I compare the same inherited 2015 rhythm under two reporting multipliers.

168 hours



orange = 10pm-2:59am

7 The data it would generate

example output structure - expected counts, not measured sounds

future hour	2015 count	baseline	multiplier	expected	scenario
2030-06-03 22:00	276	0.756	1.50	1.134	B
2030-06-03 23:00	299	0.819	1.50	1.229	B
2030-06-04 12:00	51	0.140	1.00	0.140	A

example: $299 / 365 \times 1.5 = 1.229$ expected requests at 11pm

8 What it cannot see

Decibels, duration, unreported noise, who has access to 311, repeated callers, population change, new venues, or why one sound becomes unacceptable to someone.

more rows do not necessarily mean more sound